higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

QUANTITY SURVEYING N4

(2050004)

3 December 2019 (X-paper)
09:00–13:00

This question paper consists of 6 pages and 1 addendum.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
QUANTITY SURVEYING N4
TIME: 4 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Sketches must be large, neat and fully labelled.
5. SECTION A must be done on the prescribed sheets provided.
6. SECTION B must be done in the answer book.
7. Do not use RED or GREEN ink.
8. The use of tippex is not allowed.
9. Write neatly and legibly.

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SECTION A

QUESTION 1

ADDENDUM A shows the plan view and a vertical section through the external wall of a garden shed. A plain opening has been created for entrance into the shed.

SPECIFICATIONS:

Masonry : Plaster bricks built in 1:3 cement mortar.
Plastering : Two 75 × 114 mm precast concrete lintels over plain opening.
Painting : Cement mortar bagging to internal and external walls and vertical reveals of the plain opening.
: 2 Coats p.v.a undercoat to internal and external walls and plain opening.

Measure all the work in the superstructure from the top of the concrete surface bed up to the underside of the timber wall plate. Measure all the items to create the plain opening.

NOTE: Measure strictly as indicated below.

1.1	1.1.1	Calculate the centre line of the external wall.	(2)
	1.1.2	Take off the 220 mm thick external brick wall.	(1)
1.2	1.2.1	Calculate the external and internal perimeters of the brick wall.	(4)
	1.2.2	Take off the external cement mortar bagging.	(1)
	1.2.3	Take off the internal cement mortar bagging.	(1)
1.3	1.3.1	Take off the external panting.	(1)
	1.3.2	Take off the internal painting.	(1)
1.4	1.4.1	Take off the 220 mm brick wall to create the plain opening.	(3)
	1.4.2	Take off external and internal bagging to create the plain opening.	(3)
	1.4.3	Take off the external and internal painting to create the plain opening.	(3)
1.5	1.5.1	Take off the cement mortar bagging in the reveals.	(3)
	1.5.2	Take off the painting in the reveals.	(3)
1.6	1.6.1	Take off the precast concrete lintels.	(3)
	1.6.2	Take off the painting to the horizontal concrete soffit.	(1)

[30]

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QUESTION 2

2.1 Square only the masonry items. (2½)

2.2 Abstract only the masonry items. (3½)

2.3 Bill only the masonry trade. (4)

[10]

TOTAL SECTION A: 40

SECTION B

QUESTION 3

A single rebated steel door frame suitable for a 110 mm thick wall is to be built in. From the given specifications, draw up a typical check list (measuring list) the 'taker off' will use to do the work.

Specifications:

- 2032 × 813 mm hollow core flush door
- 2-Lever ‘Union’ type lock and furniture
- 19 mm Thick plaster both sides of the wall
- Two coats PVA paint on plastered walls
- 75 mm Meranti timber skirting
- 19 mm meranti quadrant
- Two coats clear varnish on timber
- One coat undercoat and Two coats gloss enamel paint on steel. **[10]**

QUESTION 4

4.1 Briefly explain why the 'taker off' makes use of abbreviations when measuring, but not in the bills of quantities (3)

4.2 Write down in full the following abbreviations:

4.2.1 Excav.

4.2.2 S.b.

4.2.3 B.b.

4.2.4 E.o.b

4.2.5 C.m.

4.2.6 Pt.

4.2.7 w.p.

(7 × 1) (7)

[10]

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QUESTION 5

List the working up actions to be done when in the process of;

5.1 Squaring of the items measured. (5)

5.2 Abstracting of the squared items. (5)

[10]

QUESTION 6

6.1 Explain briefly what is meant by;

6.1.1 Pre-contract duties of a quantity surveyor.

6.1.2 Post-contract duties of a quantity surveyor.

(2 × 2) (4)

6.2 Name THREE pre-contract duties of a quantity surveyor. (3)

6.3 Name THREE post-contract duties of a quantity surveyor. (3)

[10]

QUESTION 7

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (7.1–7.10) in the ANSWER BOOK.

7.1 The bills of quantities are prepared by the clerk of works.

7.2 The bills of quantities are used as a basis to prepare interim valuations.

7.3 The signed bills of quantities are a legal document.

7.4 Preambles are trade specification notes in the bills of quantities.

7.5 Provisional quantities are quantities that will be re-measured as the work is carried out.

7.6 The quantity surveyor is employed by the client.

7.7 Site instructions are issued by the site foreman.

7.8 Variation orders are changes to the original contract.

7.9 Variation orders are valued by the quantity surveyor.

7.10 Contract drawings are legal contract documents.

(10 × 1) **[10]**

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QUESTION 8

Draw a neat sketch of half full bolted timber roof truss and show the following members clearly.



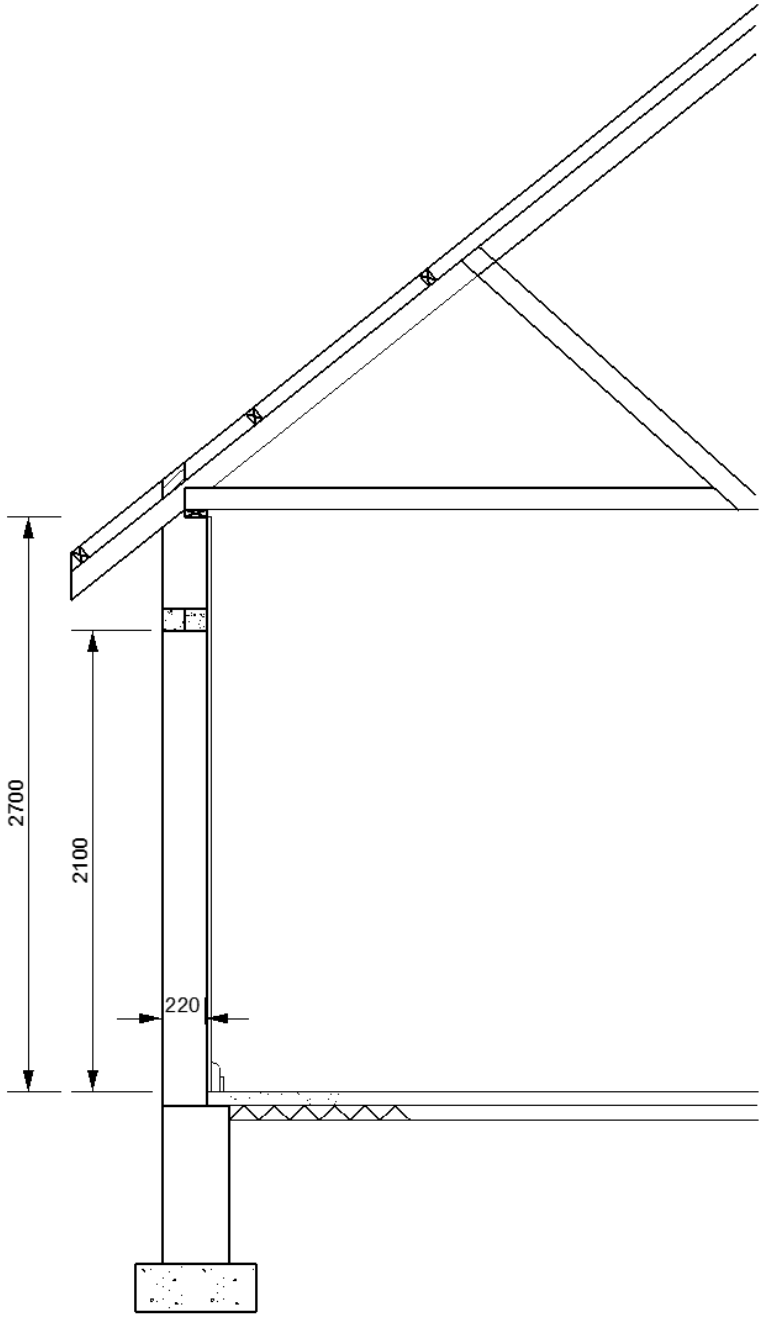
Rafter, king post, diagonal strut, tie beam, ridge **[10]**

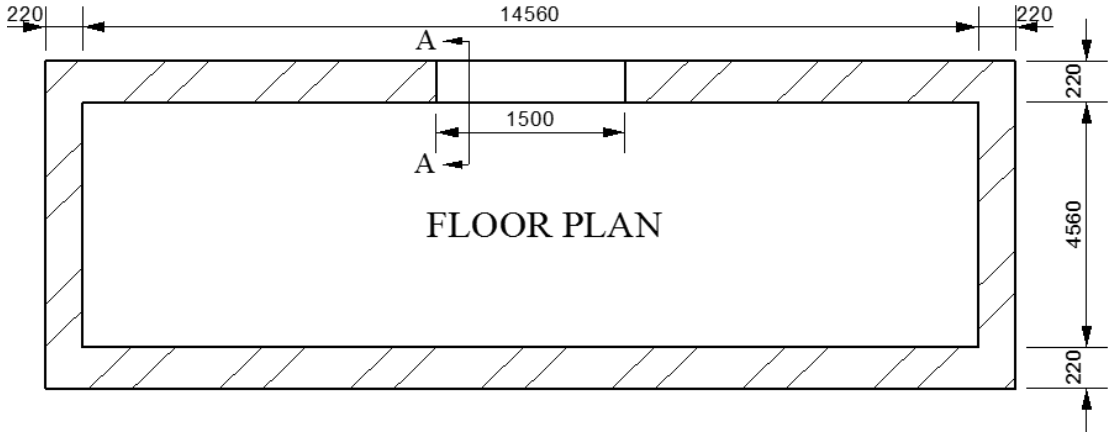
TOTAL SECTION B: 60
GRAND TOTAL: 100

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ADDENDUM A





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